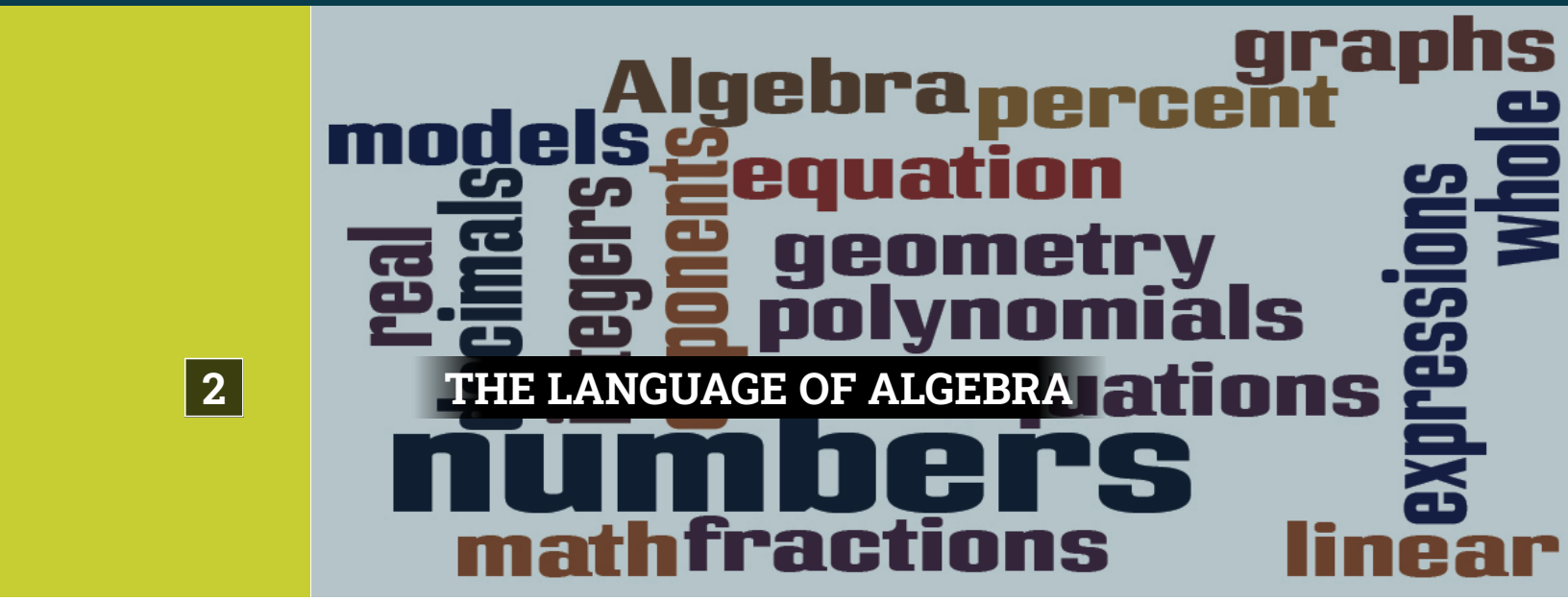


2 THE LANGUAGE OF ALGEBRA

Figure 2.1 Algebra has a language of its own. The picture shows just some of the words you may see and use in your study of Prealgebra.

Chapter Outline

2.1 Use the Language of Algebra
2.2 Evaluate, Simplify, and Translate Expressions
2.3 Solving Equations Using the Subtraction and Addition Properties of Equality
2.4 Find Multiples and Factors
2.5 Prime Factorization and the Least Common Multiple

Introduction to the Language of Algebra

You may not realize it, but you already use algebra every day. Perhaps you figure out how much to tip a server in a restaurant. Maybe you calculate the amount of change you should get when you pay for something. It could even be when you compare batting averages of your favorite players. You can describe the algebra you use in specific words, and follow an orderly process. In this chapter, you will explore the words used to describe algebra and start on your path to solving algebraic problems easily, both in class and in your everyday life.

2.1 Use the Language of Algebra

Learning Objectives

By the end of this section, you will be able to:

- Use variables and algebraic symbols
- Identify expressions and equations
- Simplify expressions with exponents
- Simplify expressions using the order of operations

BE PREPARED 2.1 Before you get started, take this readiness quiz.

Add: $43 + 69$.
If you missed this problem, review Example 1.19.

BE PREPARED 2.2 Multiply: $(896)201$.
If you missed this problem, review Example 1.48.

BE PREPARED 2.3 Divide: $7{,}263 \div 9$.

If you missed this problem, review Example 1.64.

Use Variables and Algebraic Symbols

Greg and Alex have the same birthday, but they were born in different years. This year Greg is 20 years old and Alex is 23, so Alex is 3 years older than Greg. When Greg was 12, Alex was 15. When Greg is 35, Alex will be 38. No matter what Greg's age is, Alex's age will always be 3 years more, right?

In the language of algebra, we say that Greg's age and Alex's age are variable and the three is a constant. The ages change, or vary, so age is a variable. The 3 years between them always stays the same, so the age difference is the constant.

In algebra, letters of the alphabet are used to represent variables. Suppose we call Greg's age g. Then we could use $g+3$ to represent Alex's age. See Table 2.1.

Greg's age	Alex's age
12	15
20	23
35	38
g	$g+3$

Table 2.1

Letters are used to represent variables. Letters often used for variables are x, y, a, b, and c.

Variables and Constants

A variable is a letter that represents a number or quantity whose value may change.

A constant is a number whose value always stays the same.

To write algebraically, we need some symbols as well as numbers and variables. There are several types of symbols we will be using. In Whole Numbers, we introduced the symbols for the four basic arithmetic operations: addition, subtraction, multiplication, and division. We will summarize them here, along with words we use for the operations and the result.

Operation	Notation	Say:	The result is...
Addition	$a+b$	a plus b	the sum of a and b
Subtraction	$a-b$	a minus b	the difference of a and b
Multiplication	$a\cdot b, (a)(b), (a)b, a(b)$	a times b	The product of a and b
Division	$a\div b, a/b, \frac{a}{b}, b\overline{)a}$	a divided by b	The quotient of a and b

In algebra, the cross symbol, $\times$, is not used to show multiplication because that symbol may cause confusion. Does $3xy$ mean $3\times y$ (three times y) or $3\cdot x\cdot y$ (three times x times y)? To make it clear, use • or parentheses for multiplication.

We perform these operations on two numbers. When translating from symbolic form to words, or from words to symbolic form, pay attention to the words *of* or *and* to help you find the numbers.

The *sum* ***of*** 5 ***and*** 3 means add 5 plus 3, which we write as $5+3$.

The *difference* ***of*** 9 ***and*** 2 means subtract 9 minus 2, which we write as $9 - 2$.
The *product* ***of*** 4 ***and*** 8 means multiply 4 times 8, which we can write as $4 \cdot 8$.
The *quotient* ***of*** 20 ***and*** 5 means divide 20 by 5, which we can write as $20 \div 5$.

EXAMPLE 2.1

Translate from algebra to words:

ⓐ $12 + 14$ ⓑ $(30)(5)$ ⓒ $64 \div 8$ ⓓ $x - y$

✓ **Solution**

ⓐ

$12 + 14$
12 plus 14
the sum of twelve and fourteen

ⓑ

$(30)(5)$
30 times 5
the product of thirty and five

ⓒ

$64 \div 8$
64 divided by 8
the quotient of sixty-four and eight

ⓓ

$x - y$
x minus y
the difference of x and y

▷ **TRY IT** 2.1 Translate from algebra to words.

ⓐ $18 + 11$ ⓑ $(27)(9)$ ⓒ $84 \div 7$ ⓓ $p - q$

▷ **TRY IT** 2.2 Translate from algebra to words.

ⓐ $47 - 19$ ⓑ $72 \div 9$ ⓒ $m + n$ ⓓ $(13)(7)$

When two quantities have the same value, we say they are equal and connect them with an *equal sign*.

Equality Symbol

$a = b$ is read a is equal to b

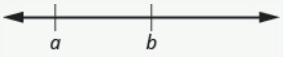
The symbol $=$ is called the equal sign.

An inequality is used in algebra to compare two quantities that may have different values. The number line can help you understand inequalities. Remember that on the number line the numbers get larger as they go from left to right. So if we know that b is greater than a, it means that b is to the right of a on the number line. We use the symbols "$<$" and "$>$" for inequalities.

Inequality

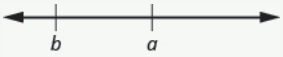
$a < b$ is read a is less than b

a is to the left of b on the number line



$a > b$ is read a is greater than b

a is to the right of b on the number line



The expressions $a < b$ and $a > b$ can be read from left-to-right or right-to-left, though in English we usually read from left-to-right. In general,

$a < b$ is equivalent to $b > a$. For example, $7 < 11$ is equivalent to $11 > 7$.

$a > b$ is equivalent to $b < a$. For example, $17 > 4$ is equivalent to $4 < 17$.

When we write an inequality symbol with a line under it, such as $a \leq b$, it means $a < b$ or $a = b$. We read this a is less than or equal to b. Also, if we put a slash through an equal sign, $\neq$, it means not equal.

We summarize the symbols of equality and inequality in Table 2.2.

Algebraic Notation	Say
$a = b$	a is equal to b
$a \neq b$	a is not equal to b
$a < b$	a is less than b
$a > b$	a is greater than b
$a \leq b$	a is less than or equal to b
$a \geq b$	a is greater than or equal to b

Table 2.2

Symbols $<$ and $>$

The symbols $<$ and $>$ each have a smaller side and a larger side.

smaller side $<$ larger side
larger side $>$ smaller side

The smaller side of the symbol faces the smaller number and the larger faces the larger number.

EXAMPLE 2.2

Translate from algebra to words:

ⓐ $20 \leq 35$ ⓑ $11 \neq 15 - 3$ ⓒ $9 > 10 \div 2$ ⓓ $x + 2 < 10$

Solution

ⓐ

$20 \leq 35$

20 is less than or equal to 35

ⓑ

$11 \neq 15 - 3$

11 is not equal to 15 minus 3

ⓒ

$9 > 10 \div 2$

9 is greater than 10 divided by 2

ⓓ

$x + 2 < 10$

x plus 2 is less than 10

TRY IT 2.3 Translate from algebra to words.

ⓐ $14 \leq 27$ ⓑ $19 - 2 \neq 8$ ⓒ $12 > 4 \div 2$ ⓓ $x - 7 < 1$

TRY IT 2.4 Translate from algebra to words.

ⓐ $19 \geq 15$ ⓑ $7 = 12 - 5$ ⓒ $15 \div 3 < 8$ ⓓ $y - 3 > 6$

EXAMPLE 2.3

The information in Figure 2.2 compares the fuel economy in miles-per-gallon (mpg) of several cars. Write the appropriate symbol $=, <, \text{or} >$ in each expression to compare the fuel economy of the cars.

Car	Prius	Mini Cooper	Toyota Corolla	Versa	Honda Fit
					
Fuel economy (mpg)	48	27	28	26	27

Figure 2.2 (credit: modification of work by Bernard Goldbach, Wikimedia Commons)

ⓐ MPG of Prius____ MPG of Mini Cooper ⓑ MPG of Versa____ MPG of Fit
ⓒ MPG of Mini Cooper____ MPG of Fit ⓓ MPG of Corolla____ MPG of Versa
ⓔ MPG of Corolla____ MPG of Prius

Solution

ⓐ

	MPG of Prius___MPG of Mini Cooper
Find the values in the chart.	48___27
Compare.	48 > 27
	MPG of Prius > MPG of Mini Cooper

ⓑ

	MPG of Versa___MPG of Fit
Find the values in the chart.	26___27
Compare.	26 < 27
	MPG of Versa < MPG of Fit

ⓒ

	MPG of Mini Cooper___MPG of Fit
Find the values in the chart.	27___27
Compare.	27 = 27
	MPG of Mini Cooper = MPG of Fit

ⓓ

	MPG of Corolla___MPG of Versa
Find the values in the chart.	28___26
Compare.	28 > 26
	MPG of Corolla > MPG of Versa

ⓔ

	MPG of Corolla___MPG of Prius
Find the values in the chart.	28___48
Compare.	28 < 48
	MPG of Corolla < MPG of Prius

TRY IT 2.5 Use Figure 2.2 to fill in the appropriate symbol, $=, <, \text{or} >$.

ⓐ MPG of Prius____MPG of Versa ⓑ MPG of Mini Cooper____ MPG of Corolla

TRY IT 2.6 Use Figure 2.2 to fill in the appropriate symbol, $=, <, \text{or} >$.

ⓐ MPG of Fit____ MPG of Prius ⓑ MPG of Corolla ____ MPG of Fit

Grouping symbols in algebra are much like the commas, colons, and other punctuation marks in written language. They indicate which expressions are to be kept together and separate from other expressions. Table 2.3 lists three of the most commonly used grouping symbols in algebra.

Common Grouping Symbols	
parentheses	( )
brackets	[ ]
braces	{ }

Table 2.3

Here are some examples of expressions that include grouping symbols. We will simplify expressions like these later in this section.

$$8(14-8) \qquad 21-3[2+4(9-8)] \qquad 24\div\{13-2[1(6-5)+4]\}$$

Identify Expressions and Equations

What is the difference in English between a phrase and a sentence? A phrase expresses a single thought that is incomplete by itself, but a sentence makes a complete statement. “Running very fast” is a phrase, but “The football player was running very fast” is a sentence. A sentence has a subject and a verb.

In algebra, we have *expressions* and *equations*. An expression is like a phrase. Here are some examples of expressions

and how they relate to word phrases:

Expression	Words	Phrase
$3+5$	3 plus 5	the sum of three and five
$n-1$	n minus one	the difference of n and one
$6 \cdot 7$	6 times 7	the product of six and seven
$\frac{x}{y}$	x divided by y	the quotient of x and y

Notice that the phrases do not form a complete sentence because the phrase does not have a verb. An equation is two expressions linked with an equal sign. When you read the words the symbols represent in an equation, you have a complete sentence in English. The equal sign gives the verb. Here are some examples of equations:

Equation	Sentence
$3+5=8$	The sum of three and five is equal to eight.
$n-1=14$	n minus one equals fourteen.
$6 \cdot 7=42$	The product of six and seven is equal to forty-two.
$x=53$	x is equal to fifty-three.
$y+9=2y-3$	y plus nine is equal to two y minus three.

Expressions and Equations

An **expression** is a number, a variable, or a combination of numbers and variables and operation symbols.

An **equation** is made up of two expressions connected by an equal sign.

EXAMPLE 2.4

Determine if each is an expression or an equation:

ⓐ $16-6=10$ ⓑ $4 \cdot 2+1$ ⓒ $x \div 25$ ⓓ $y+8=40$

Solution

ⓐ $16-6=10$	This is an equation—two expressions are connected with an equal sign.
ⓑ $4 \cdot 2+1$	This is an expression—no equal sign.
ⓒ $x \div 25$	This is an expression—no equal sign.
ⓓ $y+8=40$	This is an equation—two expressions are connected with an equal sign.

TRY IT 2.7 Determine if each is an expression or an equation:

ⓐ $23 + 6 = 29$ ⓑ $7 \cdot 3 - 7$

TRY IT 2.8 Determine if each is an expression or an equation:

$y \div 14$ $x - 6 = 21$

Simplify Expressions with Exponents

To simplify a numerical expression means to do all the math possible. For example, to simplify $4 \cdot 2 + 1$ we'd first multiply $4 \cdot 2$ to get 8 and then add the 1 to get 9. A good habit to develop is to work down the page, writing each step of the process below the previous step. The example just described would look like this:

$$4 \cdot 2 + 1$$
$$8 + 1$$
$$9$$

Suppose we have the expression $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$. We could write this more compactly using exponential notation. Exponential notation is used in algebra to represent a quantity multiplied by itself several times. We write $2 \cdot 2 \cdot 2$ as 2^3 and $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ as 2^9. In expressions such as 2^3, the 2 is called the **base** and the 3 is called the exponent. The exponent tells us how many factors of the base we have to multiply.

base → 2^3 ← exponent

$$\text{means multiply three factors of } 2$$

We say 2^3 is in exponential notation and $2 \cdot 2 \cdot 2$ is in expanded notation.

Exponential Notation

For any expression a^n, a is a factor multiplied by itself n times if n is a positive integer.

$$a^n \text{ means multiply } n \text{ factors of } a$$

base → a^n ← exponent

$$a^n = \underbrace{a \cdot a \cdot a \cdot \ldots \cdot a}_{n \text{ factors}}$$

The expression a^n is read a to the nth power.

For powers of $n = 2$ and $n = 3$, we have special names.

a^2 is read as "a squared"
a^3 is read as "a cubed"

Table 2.4 lists some examples of expressions written in exponential notation.

Exponential Notation	In Words
7^2	7 to the second power, or 7 squared
5^3	5 to the third power, or 5 cubed
9^4	9 to the fourth power
12^5	12 to the fifth power

Table 2.4

EXAMPLE 2.5

Write each expression in exponential form:

ⓐ $16 \cdot 16 \cdot 16 \cdot 16 \cdot 16 \cdot 16 \cdot 16$ ⓑ $9 \cdot 9 \cdot 9 \cdot 9 \cdot 9$ ⓒ $x \cdot x \cdot x \cdot x$ ⓓ $a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a$

Solution

ⓐ The base 16 is a factor 7 times.	16^7
ⓑ The base 9 is a factor 5 times.	9^5
ⓒ The base x is a factor 4 times.	x^4
ⓓ The base a is a factor 8 times.	a^8

TRY IT 2.9 Write each expression in exponential form:

$41 \cdot 41 \cdot 41 \cdot 41 \cdot 41$

TRY IT 2.10 Write each expression in exponential form:

$7 \cdot 7 \cdot 7 \cdot 7 \cdot 7 \cdot 7 \cdot 7 \cdot 7 \cdot 7$

EXAMPLE 2.6

Write each exponential expression in expanded form:

ⓐ 8^6 ⓑ x^5

Solution

ⓐ The base is 8 and the exponent is 6, so 8^6 means $8 \cdot 8 \cdot 8 \cdot 8 \cdot 8 \cdot 8$

ⓑ The base is x and the exponent is 5, so x^5 means $x \cdot x \cdot x \cdot x \cdot x$

TRY IT 2.11 Write each exponential expression in expanded form:

ⓐ 4^8 ⓑ a^7

TRY IT 2.12 Write each exponential expression in expanded form:

ⓐ 8^8 ⓑ b^6

To simplify an exponential expression without using a calculator, we write it in expanded form and then multiply the factors.

EXAMPLE 2.7

Simplify: 3^4.

Solution

	3^4
Expand the expression.	$3 \cdot 3 \cdot 3 \cdot 3$

Multiply left to right.	$9 \cdot 3 \cdot 3$
	$27 \cdot 3$
Multiply.	81

TRY IT 2.13 Simplify:

(a) 5^3 (b) 1^7

TRY IT 2.14 Simplify:

(a) 7^2 (b) 0^5

Simplify Expressions Using the Order of Operations

We've introduced most of the symbols and notation used in algebra, but now we need to clarify the order of operations. Otherwise, expressions may have different meanings, and they may result in different values.

For example, consider the expression:

$$4 + 3 \cdot 7$$

Some students say it simplifies to 49.

	$4 + 3 \cdot 7$
Since $4 + 3$ gives 7.	$7 \cdot 7$
And $7 \cdot 7$ is 49.	49

Some students say it simplifies to 25.

	$4 + 3 \cdot 7$
Since $3 \cdot 7$ is 21.	$4 + 21$
And $21 + 4$ makes 25.	25

Imagine the confusion that could result if every problem had several different correct answers. The same expression should give the same result. So mathematicians established some guidelines called the order of operations, which outlines the order in which parts of an expression must be simplified.

Order of Operations

When simplifying mathematical expressions perform the operations in the following order:

1. **P**arentheses and other Grouping Symbols
   - Simplify all expressions inside the parentheses or other grouping symbols, working on the innermost parentheses first.
2. **E**xponents
   - Simplify all expressions with exponents.
3. **M**ultiplication and **D**ivision
   - Perform all multiplication and division in order from left to right. These operations have equal priority.
4. **A**ddition and **S**ubtraction
   - Perform all addition and subtraction in order from left to right. These operations have equal priority.

Students often ask, "How will I remember the order?" Here is a way to help you remember: Take the first letter of each key word and substitute the silly phrase. **P**lease **E**xcuse **M**y **D**ear **A**unt **S**ally.

Order of Operations	
Please	**P**arentheses
Excuse	**E**xponents
My **D**ear	**M**ultiplication and **D**ivision
Aunt **S**ally	**A**ddition and **S**ubtraction

It's good that '**M**y **D**ear' goes together, as this reminds us that **m**ultiplication and **d**ivision have equal priority. We do not always do multiplication before division or always do division before multiplication. We do them in order from left to right.

Similarly, '**A**unt **S**ally' goes together and so reminds us that **a**ddition and **s**ubtraction also have equal priority and we do them in order from left to right.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity Game of 24 will give you practice using the order of operations.

EXAMPLE 2.8

Simplify the expressions:

ⓐ $4+3\cdot 7$ ⓑ $(4+3)\cdot 7$

Solution

ⓐ

	$4+3\cdot 7$
Are there any **p**arentheses? No.	
Are there any **e**xponents? No.	
Is there any **m**ultiplication or **d**ivision? Yes.	
Multiply first.	$4+3\cdot 7$
Add.	$4+21$
	25

ⓑ

	$(4+3)\cdot 7$
Are there any **p**arentheses? Yes.	$(4+3)\cdot 7$

Simplify inside the parentheses.	$(7)7$
Are there any **e**xponents? No.	
Is there any **m**ultiplication or **d**ivision? Yes.	
Multiply.	49

TRY IT 2.15 Simplify the expressions:

ⓐ $12 - 5 \cdot 2$ ⓑ $(12 - 5) \cdot 2$

TRY IT 2.16 Simplify the expressions:

ⓐ $8 + 3 \cdot 9$ ⓑ $(8 + 3) \cdot 9$

EXAMPLE 2.9

Simplify:

ⓐ $18 \div 9 \cdot 2$ ⓑ $18 \cdot 9 \div 2$

Solution

ⓐ

	$18 \div 9 \cdot 2$
Are there any **p**arentheses? No.	
Are there any **e**xponents? No.	
Is there any **m**ultiplication or **d**ivision? Yes.	
Multiply and divide from left to right. Divide.	$2 \cdot 2$
Multiply.	4

ⓑ

	$18 \cdot 9 \div 2$
Are there any **p**arentheses? No.	
Are there any **e**xponents? No.	
Is there any **m**ultiplication or **d**ivision? Yes.	
Multiply and divide from left to right.	

Multiply.	$162 \div 2$
Divide.	81

TRY IT 2.17 Simplify:

$42 \div 7 \cdot 3$

TRY IT 2.18 Simplify:

$12 \cdot 3 \div 4$

EXAMPLE 2.10

Simplify: $18 \div 6 + 4(5 - 2)$.

Solution

	$18 \div 6 + 4(5 - 2)$
Parentheses? Yes, subtract first.	$18 \div 6 + 4(3)$
Exponents? No.	
Multiplication or division? Yes.	
Divide first because we multiply and divide left to right.	$3 + 4(3)$
Any other multiplication or division? Yes.	
Multiply.	$3 + 12$
Any other multiplication or division? No.	
Any addition or subtraction? Yes.	15

TRY IT 2.19 Simplify:

$30 \div 5 + 10(3 - 2)$

TRY IT 2.20 Simplify:

$70 \div 10 + 4(6 - 2)$

When there are multiple grouping symbols, we simplify the innermost parentheses first and work outward.

EXAMPLE 2.11

Simplify: $5 + 2^3 + 3\left[6 - 3(4 - 2)\right]$.

Solution

	$5+2^3+3[6-3(4-2)]$
Are there any parentheses (or other grouping symbol)? Yes.	
Focus on the parentheses that are inside the brackets.	$5+2^3+3[6-3(4-2)]$
Subtract.	$5+2^3+3[6-3(2)]$
Continue inside the brackets and multiply.	$5+2^3+3[6-6]$
Continue inside the brackets and subtract.	$5+2^3+3[0]$
The expression inside the brackets requires no further simplification.	
Are there any exponents? Yes.	
Simplify exponents.	$5+2^3+3[0]$
Is there any multiplication or division? Yes.	
Multiply.	$5+8+3[0]$
Is there any addition or subtraction? Yes.	
Add.	$5+8+0$
Add.	$13+0$
	13

TRY IT 2.21 Simplify:

$9+5^3-[4(9+3)]$

TRY IT 2.22 Simplify:

$7^2-2[4(5+1)]$

EXAMPLE 2.12

Simplify: $2^3+3^4\div 3-5^2$.

Solution

	$2^3+3^4\div 3-5^2$
If an expression has several exponents, they may be simplified in the same step.	

Simplify exponents.	$2^3 + 3^4 \div 3 - 5^2$
Divide.	$8 + 81 \div 3 - 25$
Add.	$8 + 27 - 25$
Subtract.	$35 - 25$
	10

TRY IT 2.23 Simplify:

$3^2 + 2^4 \div 2 + 4^3$

TRY IT 2.24 Simplify:

$6^2 - 5^3 \div 5 + 8^2$

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Order of Operations (https://openstaxcollege.org/l/24orderoperate)

Order of Operations – The Basics (https://openstaxcollege.org/l/24orderbasic)

Ex: Evaluate an Expression Using the Order of Operations (https://openstaxcollege.org/l/24Evalexpress)

Example 3: Evaluate an Expression Using The Order of Operations (https://openstaxcollege.org/l/24evalexpress3)

SECTION 2.1 EXERCISES

Practice Makes Perfect

Use Variables and Algebraic Symbols

In the following exercises, translate from algebraic notation to words.

1. $16 - 9$

2. $25 - 7$

3. $5 \cdot 6$

4. $3 \cdot 9$

5. $28 \div 4$

6. $45 \div 5$

7. $x + 8$

8. $x + 11$

9. $(2)(7)$

10. $(4)(8)$

11. $14 < 21$

12. $17 < 35$

13. $36 \geq 19$

14. $42 \geq 27$

15. $3n = 24$

16. $6n = 36$

17. $y - 1 > 6$

18. $y - 4 > 8$

19. $2 \leq 18 \div 6$

20. $3 \leq 20 \div 4$

21. $a \neq 7 \cdot 4$

22. $a \neq 1 \cdot 12$

Identify Expressions and Equations

In the following exercises, determine if each is an expression or an equation.

23. $9 \cdot 6 = 54$

24. $7 \cdot 9 = 63$

25. $5 \cdot 4 + 3$

26. $6 \cdot 3 + 5$

27. $x + 7$

28. $x + 9$

29. $y - 5 = 25$

30. $y - 8 = 32$

Simplify Expressions with Exponents

In the following exercises, write in exponential form.

31. $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$

32. $4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$

33. $x \cdot x \cdot x \cdot x \cdot x$

34. $y \cdot y \cdot y \cdot y \cdot y \cdot y$

In the following exercises, write in expanded form.

35. 5^3

36. 8^3

37. 2^8

38. 10^5

Simplify Expressions Using the Order of Operations

In the following exercises, simplify.

39. ⓐ $3 + 8 \cdot 5$ ⓑ $(3+8) \cdot 5$

40. ⓐ $2 + 6 \cdot 3$ ⓑ $(2+6) \cdot 3$

41. $2^3 - 12 \div (9 - 5)$

42. $3^2 - 18 \div (11 - 5)$

43. $3 \cdot 8 + 5 \cdot 2$

44. $4 \cdot 7 + 3 \cdot 5$

45. $2 + 8(6 + 1)$

46. $4 + 6(3 + 6)$

47. $4 \cdot 12/8$

48. $2 \cdot 36/6$

49. $6 + 10/2 + 2$

50. $9 + 12/3 + 4$

51. $(6 + 10) \div (2 + 2)$

52. $(9 + 12) \div (3 + 4)$

53. $20 \div 4 + 6 \cdot 5$

54. $33 \div 3 + 8 \cdot 2$

55. $20 \div (4 + 6) \cdot 5$

56. $33 \div (3 + 8) \cdot 2$

57. $4^2 + 5^2$

58. $3^2 + 7^2$

59. $(4 + 5)^2$

60. $(3 + 7)^2$

61. $3(1 + 9 \cdot 6) - 4^2$

62. $5(2 + 8 \cdot 4) - 7^2$

63. $2[1 + 3(10 - 2)]$

64. $5[2 + 4(3 - 2)]$

Everyday Math

65. **Basketball** In the 2014 NBA playoffs, the San Antonio Spurs beat the Miami Heat. The table below shows the heights of the starters on each team. Use this table to fill in the appropriate symbol $(=, <, >)$.

Spurs	Height		Heat	Height
Tim Duncan	$83''$		Rashard Lewis	$82''$
Boris Diaw	$80''$		LeBron James	$80''$
Kawhi Leonard	$79''$		Chris Bosh	$83''$
Tony Parker	$74''$		Dwyane Wade	$76''$
Danny Green	$78''$		Ray Allen	$77''$

ⓐ Height of Tim Duncan___Height of Rashard Lewis
ⓑ Height of Boris Diaw___Height of LeBron James
ⓒ Height of Kawhi Leonard___Height of Chris Bosh
ⓓ Height of Tony Parker___Height of Dwyane Wade
ⓔ Height of Danny Green___Height of Ray Allen

66. **Elevation** In Colorado there are more than 50 mountains with an elevation of over $14{,}000$ feet. The table shows the ten tallest. Use this table to fill in the appropriate inequality symbol.

Mountain	Elevation
Mt. Elbert	$14{,}433'$
Mt. Massive	$14{,}421'$
Mt. Harvard	$14{,}420'$
Blanca Peak	$14{,}345'$
La Plata Peak	$14{,}336'$
Uncompahgre Peak	$14{,}309'$
Crestone Peak	$14{,}294'$
Mt. Lincoln	$14{,}286'$
Grays Peak	$14{,}270'$
Mt. Antero	$14{,}269'$

ⓐ Elevation of La Plata Peak___Elevation of Mt. Antero
ⓑ Elevation of Blanca Peak___Elevation of Mt. Elbert
ⓒ Elevation of Gray's Peak___Elevation of Mt. Lincoln
ⓓ Elevation of Mt. Massive___Elevation of Crestone Peak
ⓔ Elevation of Mt. Harvard___Elevation of Uncompahgre Peak

Writing Exercises

67. Explain the difference between an expression and an equation.

68. Why is it important to use the order of operations to simplify an expression?